

Appendices

Appendix: Pipeline Stage Reference I

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Table 1: Single-project DAG exported from default pipeline.yaml (names shown in topological order). Scripts live under scripts/.

Stage name	Script / method	Primary inputs	Outputs / artefacts	Failure mode
Clean Output Directories	_run_clean_outputs	prior projects/<name>/output/, mirrored output/<name>/	emptied trees	Blocking
Environment Setup	scripts/pipeline/stage_00_setup.py	toolchain probes	scaffold dirs, env exports	Blocking
Infrastructure Tests	scripts/pipeline/stage_01_infra.py --infra-only --infra-scope pipeline-smoke	tests/infra/	coverage + junit-style logs	tolerant ceilings
Project Tests	scripts/pipeline/stage_02_project.py --project-only	projects/<name>/tests/	stage artefacts	blocking by default
Project Analysis	scripts/pipeline/stage_03_analysis.py	thin scripts	figures/, data/, reports	Blocking
Connector Search	scripts/pipeline/stage_04_connector_search.py ge_08_connector_search.yaml	manuscript/config	output/data/connector_search/	opt-in (science tag); skipped if not configured
Provenance Record	scripts/pipeline/stage_05_provenance.py --stage "Connector Search"	prior stage endpoints	provenance/graph.json	opt-in (provenance tag); skipped if not configured
PDF Rendering	scripts/pipeline/stage_06_pdf_render.py ge_03_render.py	manuscript/, placeholders	.pdf/.tex bundles	Blocking
Output Validation	scripts/pipeline/stage_07_validation.py ge_04_validate.py	render tree	Markdown + PDF diagnostics	Blocking / downgrade
LLM Scientific Review	scripts/pipeline/stage_08_llm_review.py --reviews-only	resolved manuscript artefacts	textual reviews	Optional skip (allow_skip)

Appendix: Pipeline Stage Reference II

LLM	scripts/pipeline/stage_06_llm_metadata.py	multilingual snippets	Optional skip (allow_skip)
Translations	--translations-only		soft fail logged
Copy Outputs	scripts/pipeline/stage_05_copy.py	validated tree	mirrored output/<name>/...
Ebook	scripts/pipeline/stage_11_ebook.py	rendered combined	output/ebook/
Generation	ge_11_ebook.py	markdown	(EPUB/MOBI/DOCX)
docxplus	scripts/pipeline/stage_13_docxplus.py	project source tree +	output/docxplus/ (.docx,
Export	ge_13_docxplus.py	metadata	.docxplus)
Metadata	scripts/pipeline/stage_12_metadata.py	manuscript/config	output/metadata/
Package	ge_12_metadata.py	.yaml	(ONIX/JSON/OPF)
Executable	scripts/runner/bundle_executable.py	project tree +	container bundle manifest
Bundle	e_executable.py	outputs	
Archival	scripts/runner/archival_publication.py	bundle + deliverables	archival deposit manifest
Publication			opt-in (archival tag)

Appendix: Pipeline Stage Reference I

`scripts/pipeline/stage_07_executive_report.py` is invoked **outside** this DAG whenever `execute_multi_project.py` aggregates pipelines—supplying cross-project KPI dashboards absent from lone-project checkpoints.